Exotic Interest Rate Swaps: **Snowballs in Portugal**

METRO DO PORTO CASE ANALYSIS

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EXECUTIVE SUMMARY

Case at a Glance

1

Metro do Porto (MdP)

Portugal's largest light rail network in Porto, burdened with **EUR 300M+** in variable-rate debt tied to **Euribor 6M** — a ticking time bomb in a volatile rate environment.

2

Two Swap Agreements (2007)

A **vanilla fixed-to-floating swap** with BCP and an exotic **"snowball" swap** with Santander Totta (**EUR 89M** notional) — vastly different in risk profile.

3

Post-Crisis Catastrophe

After the **2008 financial crisis**, Euribor plunged, triggering the snowball clause — MdP's coupon obligations spiraled beyond **60% per annum**, far exceeding any conceivable hedge benefit.

4

Fundamental Questions Raised

The case raises critical questions about **hedging vs. speculation**, derivative complexity, and **public sector risk governance** — with implications far beyond Portugal.

CASE SETTING

Metro do Porto: A Public Transit Giant Under Pressure

Company Profile

- **State-owned light rail system** in Porto, Portugal — a critical public infrastructure asset
- Serves Portugal's **second-largest** metropolitan area with over 1.2 million residents
- Critical infrastructure for regional economic development and citizen mobility
- Heavily dependent on **government grants & subsidies** to fund operations and service debt

Financial Distress Indicators

- **EUR 300M+** in variable-rate debt linked to Euribor 6M — massive interest rate exposure
- Rising interest expenses consuming operating cash flows, crowding out capital investment
- **Negative net income** despite growing ridership — structural profitability gap
- Leverage ratios with Debt/Equity exceeding **5x** — extremely high for a public entity
- Persistent **negative operating cash flows** — unable to self-fund without external support

€300M+	VARIABLE-RATE DEBT EXPOSURE
5x>	DEBT / EQUITY LEVERAGE RATIO
Negative	NET INCOME & OPERATING CASH FLOW
6M	EURIBOR TENOR RATE BENCHMARK

What are the payoffs, potential profits and losses, benefits, and risks of the plain-vanilla swap implemented with BCP? Would you qualify this transaction as hedging?

Swap 1: Vanilla Interest Rate Swap with BCP

BCP
Swap Counterparty

Pays Fixed
4.76%

Receives
Euribor 6M

SWAP AGREEMENT

Metro do Porto
MdP — Swap Client

Bondholders
Existing Debt Holders

Pays
Euribor 6M + Spread

EXISTING DEBT

WITHOUT SWAP
Euribor 6M + Spread
(Variable & uncertain)

WITH SWAP
4.76% Spread
(Fixed & predictable)

KEY TERMS

Notional	~EUR 96.2M
Tenor	20 Years (Maturity: Dec 2022)
Payment Frequency	Semi-Annually
Fixed Rate (MdP pays)	4.76%
Floating Rate (MdP receives)	Euribor 6M

Initial Status (2006):

Unprofitable. Average Euribor was only 2.47%, leading to a **€17M cash outflow to BCP**.

Vanilla Swap — Scenario Analysis

MdP fixed rate: 4.76% | 20-year tenor | ~EUR 96.2M notional | Semi-annual payments

METRIC	SCENARIO A	SCENARIO B	SCENARIO C
Euribor 6M Path	3.50% stable	6.00% gradual rise	1.00% sharp decline
MdP Pays Fixed	4.76% to BCP	4.76% to BCP	4.76% to BCP
MdP Receives	Euribor 6M ≈ 3.50%	Euribor 6M avg ≈ 6.00%+	Euribor 6M ≈ 1.00%
Net Swap Cost	+1.26% Slight cost for certainty	−1.24% (Net Gain) Significant savings	+3.76% Higher cost, but predictable
Outcome	Neutral/Predictable	Excellent Hedge	Costly but Bounded

VERDICT

The vanilla swap **IS a legitimate hedge** — it provides certainty of funding costs. MdP gives up potential savings if rates fall in exchange for protection if rates rise. All outcomes are bounded and predictable.

What are the payoffs, potential profits and losses, benefits and risks
of the exotic swap proposed by Santander?
Would you qualify this
transaction as hedging?

SWAP STRUCTURE

Swap 2: The Snowball Swap **with Santander Totta**

Phase 1 · Years 1–2 · Quarters 1–8

Teaser Period

MdP → 4.13% Fixed → Santander

Santander → Euribor 6M → MdP

Attractive below-market rate — designed to entice MdP into the structure

Phase 2 · Years 3–10 · Quarters 9–40

Snowball Coupon Activates

THE SNOWBALL FORMULA

$$C(t) = \max(0\%, C(t-1) + 4.50\% - 2.0 \times \text{Euribor}_{6M})$$

- **Path dependency:** Previous coupon C(t−1) carries forward into the next period
- **Trigger point:** When Euribor falls below **2.25%**, the coupon *increases* each quarter
- **Snowball effect:** Accumulation is exponential — like a snowball rolling downhill

KEY TERMS

Notional	EUR 89M
Tenor	10 Years
Payments	Quarterly
Floor	0% (min coupon)
Teaser Rate	~4.13% Fixed
Snowball K	4.50%
Multiplier	2.0 × Euribor

CRITICAL THRESHOLD

2.25%

Euribor Breakeven

Above 2.25%: Coupon decreases → falls to 0% floor → favorable for MdP

Below 2.25%: Coupon increases every quarter → compounds relentlessly → catastrophic

DEEP DIVE

Anatomy of the Snowball Mechanism

SNOWBALL FORMULA

$$C(t) = \max(0\%,\ C(t-1) + 4.50\% - 2.0 \times \text{Euribor6M})$$

Assuming Euribor 6M = **1.0%** → each quarter adds **+2.50%**

STEP-BY-STEP ACCUMULATION

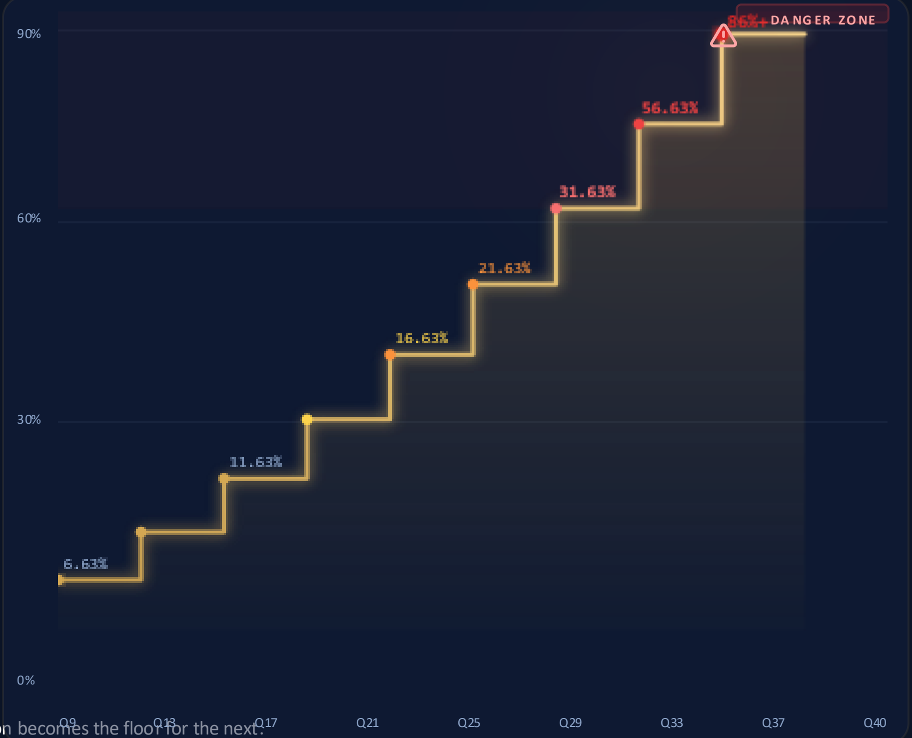
Q9	max(0%, 4.13% + 4.50% − 2.0%)	6.63%
Q10	max(0%, 6.63% + 4.50% − 2.0%)	9.13%
Q11	max(0%, 9.13% + 4.50% − 2.0%)	11.63%
Q12	max(0%, 11.63% + 4.50% − 2.0%)	14.13%
Q13	max(0%, 14.13% + 4.50% − 2.0%)	16.63%

. . .

AFTER 10 QUARTERS	AFTER 20 QUARTERS	AFTER 32 QUARTERS
31.63%	56.63%	86.63%

"SNOWBALL" — small initial differences compound into enormous obligations. Each quarter's coupon becomes the floor for the next.

COUPON ESCALATION STAIRCASE



Path dependency is the critical risk — each quarter's coupon *carries forward*, making the snowball impossible to reverse once triggered.

✓ SCENARIO A

Snowball Swap — Stable Rates (~3.70%)

EURIBOR 6M
3.70%
held constant

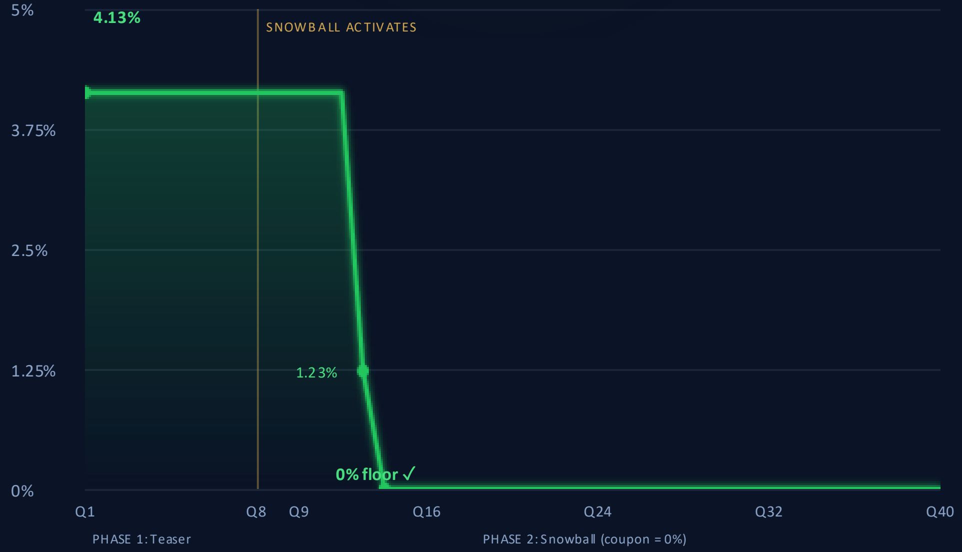
SNOWBALL FORMULA AT 3.70% EURIBOR

```
C(t) = max(0%, C(t-1) + 4.50% − 2 × 3.70%)

             = max(0%, C(t-1) − 2.90%)
```

Because 2 × 3.70% = 7.40% > 4.50%, the coupon **decreases** each quarter by 2.90%

MDP COUPON RATE OVER 40 QUARTERS



QUARTERLY COUPON WALKTHROUGH

Q1–8	Teaser rate (Phase 1)	4.13%
Q9	max(0, 4.13 − 2.90)	1.23%
Q10	max(0, 1.23 − 2.90)	0.00%
Q11	max(0, 0.00 − 2.90)	0.00%
Q12	max(0, 0.00 − 2.90)	0.00%
	. . .	
Q40	max(0, 0.00 − 2.90)	0.00%

The coupon hits the **0% floor** by Quarter 10 and stays there for **30+ quarters**. MdP pays nothing on the snowball leg.

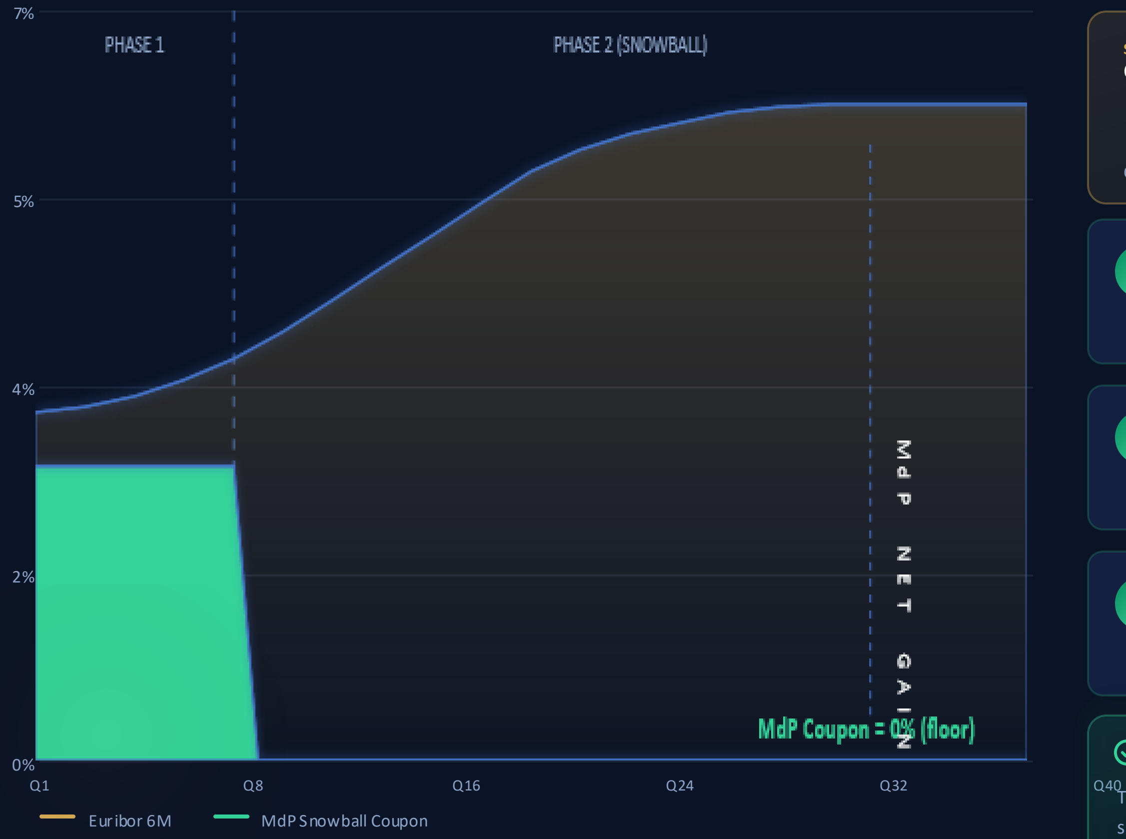
NET EFFECT FOR MDP

MdP Pays	MdP Receives
0%	3.70%
snowball leg	Euribor 6M

SCENARIO B

BEST CASE FOR MdP

Snowball Swap — Rising Rates (to 6%)



SNOWBALL FORMULA

```
C(t) = max(0%, C(t-1) + 4.50% - 2 × 6.0%)

     = max(0%, C(t-1) - 8.50%)
```

Coupon drops by 9.50% each quarter → hits 0% floor instantly

MdP Pays Zero

Snowball coupon plummets to **0% floor** immediately in Phase 2. MdP owes nothing on the snowball leg.

MdP Receives High Euribor

Santander pays MdP **Euribor 6M (≈6%)** on €89M notional — substantial quarterly cash inflows.

Net Gain: ~€6.2M / Year

Receiving 6% on €89M = **€6.23M annually** while paying zero. Maximum upside for MdP.

VERDICT: BEST CASE

This is the scenario MdP was betting on. With rising or stable rates, the snowball swap is **extremely favorable** — unlimited Euribor receipts with zero coupon payments. But rates didn't rise.

SCENARIO C — CATASTROPHE

Snowball Swap — Falling Rates (to 1%)

When Euribor drops to 1%, each quarter adds +2.50% to the coupon — compounding into catastrophic obligations.

SNOWBALL FORMULA

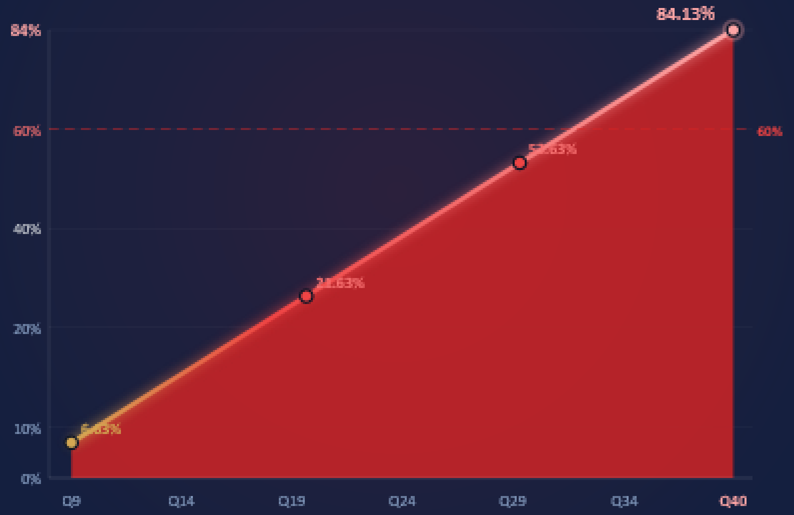
C(t) = max(0%, C(t-1) + 4.50% – 2 × 1.0%)

= C(t-1) + **2.50%** every quarter

COUPON ESCALATION

QUARTER	COUPON	QTR PAYMENT
Q9	6.63%	€1.48M
Q10	9.13%	€2.03M
Q11	11.63%	€2.59M
Q12	14.13%	€3.14M
...		
Q20	34.13%	€7.59M
...		
Q30	59.13%	€13.16M
...		
Q40	**84.13%**	**€18.72M**

ANNUALIZED COUPON RATE OVER TIME



€13.35M
PER QUARTER AT 60%

€53M+
ANNUALIZED AT 60%

€300M+
TOTAL CUMULATIVE EXPOSURE

€89M Notional
LOSSES EXCEED 3× NOTIONAL

This is what ***actually happened*** to Metro do Porto after the 2008 financial crisis crushed Euribor rates.

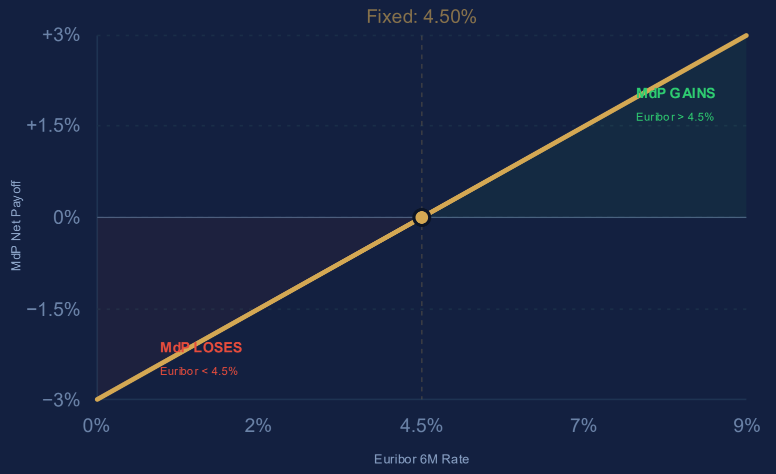
The snowball effect in action: A 2.50% quarterly increment on €89M notional transforms modest initial payments into obligations exceeding MdP's entire annual revenue — an exponential spiral with no natural ceiling.

Should Metro do Porto implement the derivative transaction proposed by Santander?

Payoff Comparison: Vanilla vs. Snowball

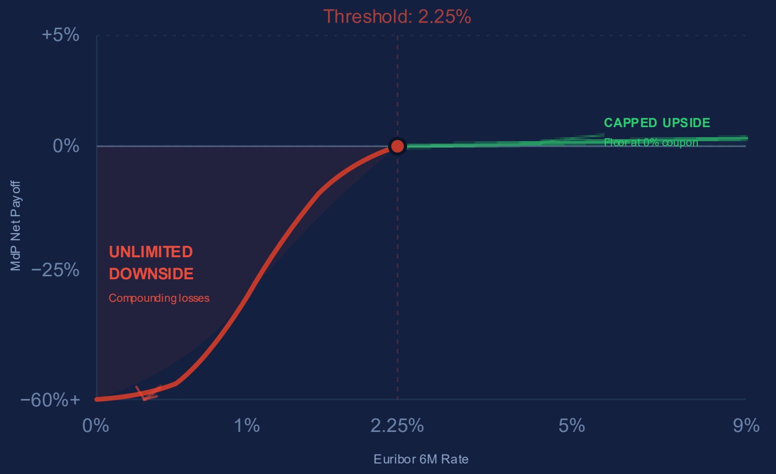
Side-by-side risk profiles reveal a fundamentally broken risk-reward structure

Vanilla Swap Payoff



LINEAR Proportional | **SYMMETRIC** Balanced risk | **BOUNDED** Predictable

Snowball Swap Payoff



EXPONENTIAL Compounding | **ASYMMETRIC** One-sided risk | **UNBOUNDED** No loss cap

Key Insight: The snowball swap provides **LIMITED upside** (coupon floored at 0%) but **UNLIMITED, COMPOUNDING downside** — a fundamentally asymmetric risk profile.

MAX GAIN ~€20M

MAX LOSS ∞

Upside vs. Downside — The Asymmetric Bet

Snowball swap risk-reward analysis on EUR 89M notional

MAXIMUM UPSIDE

Teaser rate savings in Years 1–2 (~4.13% vs market)

Coupon drops to **0% floor** if rates stay above 2.25%

MdP receives Euribor 6M from Santander

~€15–20M TOTAL UPSIDE CAPPED OVER 10 YEARS

Relative scale of potential gain

VS

UNLIMITED DOWNSIDE

No cap on losses — compounding snowball effect

Losses can exceed **€200–300M+**

At 60%+ annualized coupon on €89M notional:

€53M+ PER YEAR IN PAYMENTS

Relative scale of potential loss

NATURE OF RISK

Path-dependent, tail-risk exposure

MdP essentially **sold a complex interest rate option**

Massive **negative convexity** — small rate moves create outsized losses

Cannot be unwound without crystallizing large MTM losses

SCALE OF EXPOSURE

Worst-case loss **exceeds MdP's entire annual revenue**

Potential loss is **multiples of the notional** amount (€89M)

Risk dwarfs the original interest rate savings objective

Cumulative exposure: **3–4x notional** in extreme scenario

Risk-Reward Ratio: Fundamentally Broken

MdP risked **catastrophic, unlimited losses (€200M+)** for **modest, capped savings (~€15–20M)** — an asymmetry no rational hedger would accept.

Is This Swap Excessively Complex?

✕ EXCESSIVELY COMPLEX

- **Path dependency** makes valuation nearly impossible without Monte Carlo simulation
- **Public sector entity** lacks sophisticated derivatives expertise
- **Embedded option exposure** is opaque — MdP effectively wrote a complex interest rate option
- The **"snowball" accumulation** mechanism is non-intuitive and obscures true risk
- **No standard risk metrics** (VaR, DV01) can adequately capture the tail risk
- Violates the **KISS principle** for public finance

VS

✓ COUNTERARGUMENTS

- Structured products are **standard** in institutional markets
- MdP had **access to financial advisors** who could evaluate the structure
- The formula itself is **mechanically simple** — it's the implications that are complex
- **Similar structures** were widely used across European municipalities

COMPLEXITY ASSESSMENT

Simple — Extreme

85% complexity score

VERDICT

The swap is **EXCESSIVELY complex** for a public sector entity. Complexity served the **bank's interests** (fees, proprietary positioning) — not the client's hedging needs.

Question 6: Hedging or Speculation?

THREE TESTS FOR HEDGING

Test 1: Does it reduce risk? NO

The snowball swap **INCREASED** MdP's risk by introducing path-dependent, unlimited downside exposure. Risk was amplified, not reduced.

Test 2: Does it offset the underlying exposure? PARTIAL

While MdP had floating-rate debt, the snowball coupon does **not move proportionally** with the underlying exposure. The non-linear, path-dependent nature breaks the hedge relationship.

Test 3: Is the notional size appropriate? YES

EUR 89M notional vs EUR 300M+ in debt is reasonable in size. The notional alone is not the problem — the structure is.

Hedging Score: **1 of 3 passed → NOT a hedge**

MdP was effectively betting that
Euribor would remain above ~2.25%

VERDICT

The snowball swap is **SPECULATION**, not hedging. While the vanilla swap with BCP was a legitimate hedge, the snowball swap transformed MdP from a hedger into a **leveraged speculator** on interest rates.

Political, Economic, & Governance Implications

Structural lessons from the Metro do Porto derivatives failure

1 Misaligned Incentives

Banks earn fees from structuring complexity; public entities lack the expertise to evaluate exotic products. The **information asymmetry** systematically favors the structuring bank, creating a principal-agent problem at the taxpayer's expense.

2 Moral Hazard

MdP management pursued **short-term interest savings** via teaser rates to improve near-term financials — at the cost of catastrophic tail risk. Decision-makers bore none of the downside personally, while taxpayers absorbed all losses.

3 Regulatory Gaps

No regulatory framework existed in Portugal — or most of Europe — to govern municipal derivatives usage. Post-crisis, ISDA documentation reforms and EU regulations (EMIR) addressed some gaps, but enforcement remains uneven.

4 Public Accountability

Taxpayers ultimately bear the cost of failed derivatives strategies by state-owned entities. Democratic oversight of financial risk-taking is essential — elected officials must understand and authorize complex financial commitments.

Core Insight: The MdP case is not merely a story of bad luck — it reveals systemic governance failures where complexity was rewarded, oversight was absent, and the public bore asymmetric risk they never understood or approved.

What would be alternatives for Metro do Porto to solve its financial problems?

Strategic Recommendations

1 Immediate Restructuring

MdP should negotiate an early termination or restructuring of the snowball swap, even at significant cost, to stop the bleeding.

2 Vanilla Hedging Only

Public entities should restrict themselves to plain-vanilla derivatives (standard IRS, caps, floors) with transparent, bounded risk profiles.

3 Independent Valuation

All structured derivative proposals must be independently valued by a third-party advisor not compensated by the structuring bank.

4 Risk Limits Framework

Implement maximum loss thresholds, mark-to-market monitoring, and derivative notional limits relative to underlying debt.

5 Board-Level Oversight

Derivative transactions above a threshold must require board approval with full risk disclosure and scenario analysis.

6 Regulatory Reform

Portugal should implement municipal derivative regulations similar to those enacted in Italy and the UK post-crisis.

From crisis management to structural governance reform

KEY TAKEAWAY

"The snowball swap transformed a public utility's legitimate need for interest rate certainty into a leveraged, one-sided bet against historically unprecedented monetary policy easing. The lesson: financial innovation without commensurate risk governance is speculation dressed as hedging."

Thank You